Class-Wise Distribution Adaptation for Unsupervised Classification of Hyperspectral Remote Sensing Images

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***Abstract*— Class-wise adversarial adaptation networks are investigated for the classification of hyperspectral remote sensing images in this article. By adversarial learning between the feature extractor and the multiple domain discriminators, domain-invariant features are generated. Moreover, a probability-prediction-based maximum mean discrepancy (MMD) method is introduced to the adversarial adaptation network to achieve a superior feature-alignment performance. The class-wise adversarial adaptation in conjunction with the class-wise probability MMD is denoted as the class-wise distribution adaptation (CDA) network. The proposed CDA does not require labeled information in the target domain and can achieve an unsupervised classification of the target image. The experimental results using the Hyperion and Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) hyperspectral data demonstrated its efficiency.**

***Index Terms*— Adversarial learning, classification, domain adaptation, remote sensing.**

I. Introduction

HYPERSPECTRAL remote sensing images have provided excellent capability for feature extraction [1], [2] and classification [3], [4]. It is acknowledged that labeling the remote sensing data is very expensive and time-consuming. Traditional classification methods require that the training data and the testing data are independently and identically distributed. However, due to the changed illumination conditions, vegetation composition, topography, and solar incidence angle [5], [6], the spectral shift exists between the multitemporal images or the spatially disjoint images. Therefore, if the training data and the testing data are from different but related images, traditional classification approaches may not achieve satisfactory performance. Fortunately, domain adaptation is able to solve this problem, since it can transfer knowledge from a source image to improve the classification of a target image. In the domain adaptation situation, the image with sufficient labels is called the source domain and the image with few labels or without labels is referred to as the target domain.

Traditional domain adaptation algorithms contain three categories: instance-based methods, classifier-based methods, and feature-based methods. The first category attempts to decrease the domain discrepancy by reweighting the data instances [7]–[9]. The second one aims to learn an adaptive classifier for target data by leveraging labeled samples from the source domain and possible few labeled samples from the target domain. The feature-based methods achieve domain adaptation by extracting the common feature representations by feature transformation [10]–[15] or feature reconstruction [16], [17].

In recent years, deep neural networks have been exploited for domain adaptation due to its powerful feature representation ability. A deep transfer network is a classifier-based technique that often cooperates with the feature-based domain adaptation strategies. Long *et al.* [18] proposed a deep adaptation network (DAN) by embedding multiple-kernel maximum mean discrepancy (MKMMD) into the network. Similarly, Sun and Saenko [19] used correlation alignment for domain adaptation (D-CORAL) by aligning the covariance matrix of the features extracted by the deep networks. Deep network-based domain adaptation approaches have also been successfully applied for the classification of remote sensing images. Ma and Song [20] achieved unsupervised domain adaptation by aligning a class centroid in a deep neural network. Elshamli *et al.* [21] used denoising autoencoders to learn domain-invariant representations. Song *et al.* [22] designed a subspace alignment and convolutional neural network-based framework to realize domain adaptation in the remote sensing scene images. Wang *et al.* [23] proposed a domain adaptation method by learning the manifold embedding and discriminative features with deep neural networks for hyperspectral image classification.

In a deep learning community, a generative adversarial network (GAN) has achieved outstanding performance in

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various fields [24]–[27]. It is able to generate real-like fake data by adversarial learning between a generator and a discriminator. Adversarial learning is highly useful for transforming images to another specific domain [28], and thus, it is suitable for domain adaptation that aims to match the two domains. Applying adversarial learning for domain adaptation results in an adversarial adaptation network, where the discriminator performs a binary domain classification, while the feature extractor generates domain-invariant features such that they cannot be separated by the discriminator. Bousmalis *et al.* [29] employed the original GAN framework to generate fake target data from the source data to acquire the relations between the domains. Tzeng *et al.* [30] used a GAN to transform the data of the two domains and make their features similar and discriminative. Ganin *et al.* [31] proposed a domain adversarial neural network (DANN) for feature alignment by maximizing the domain classification loss. Shen *et al.* [32] took advantage of the Wasserstein distance to measure the distribution divergency between the domains to learn the domain-invariant features. Pei *et al.* [33] proposed a multiple adversarial domain adaptation network (MADA) to achieve distribution adaptation by using multiple domain discriminators.

Adversarial adaptation has also been applied for the classification of remote sensing images. Elshamli *et al.* [21] used an adversarial adaptation network to achieve domain-invariant features for the classification of remote sensing data. Zhang *et al.* [34] applied the adversarial adaptation network for the classification of multiband Synthetic Aperture Radar (SAR) images. Bejiga and Melgani [35], [36] applied the GANs in the context of aerial image classification and cross-sensor hyperspectral data classification.

In this article, we explore the application of the adversarial adaptation for the unsupervised classification of the hyperspectral remote sensing images. The existing adversarial adaptation networks for the classification of remote sensing data align only the marginal distributions across the domains in the feature space [21], [34], [35], which cannot guarantee that their class-conditional distributions are also drawn close. For remote sensing images, different classes may have different spectral shifts. Aligning the marginal distribution between the domains is not equivalent to aligning their class-conditional distributions. Therefore, the adversarial adaptation method should be generalized to align the features of each specific class. The adversarial adaptation conducted on each class is called class-wise adversarial adaptation in this article. To the best of our knowledge, the class-wise adversarial adaptation has not been applied for the classification of hyperspectral remote sensing images.

Class-wise adversarial adaptation requires labeled data from each class. Since there is no labeled information in the target domain, the predicted labels of the target data are used instead. However, due to the spectral drift, classification accuracy without domain adaptation may be very low, resulting in an inferior adaptation performance. To mitigate this problem, we introduce another domain adaptation strategy, i.e., maximum mean discrepancy (MMD), into the adversarial adaptation network. Considering the adaptation of conditional distribution, the MMD is conducted on a per-class basis. The class-wise MMD calculates the first-order statistics, which is more robust to the false pseudolabels than the class-wise adversarial adaptation strategy. Moreover, since soft labels contain more information than hard labels, we use the predicted probability outputs of the target data to estimate the centroid of each class in the MMD, which is called the probability MMD (PMMD) method. It is expected that a combination of adversarial adaptation and PMMD can result in a superior adaptation performance.

The proposed transfer network is able to conduct feature extraction and classification simultaneously. The classifier is trained by both the classification loss of the source labeled data and an entropy constraint on the data from the target domain. This transfer network achieves distribution adaptation with both the class-wise adversarial adaptation and class-wise PMMD strategy, and thus is named the class-wise distribution adaptation (CDA) network. The proposed CDA network has the following properties.

1) It considers the class-specific relationships between the source domain and the target domain and can extract the domain-invariant features on a per-class basis.
2) It combines both the adversarial adaptation strategy and the PMMD strategy to yield better feature alignment.
3) Target labels are not required, and unsupervised classification can be achieved.

The organization of this article is as follows. Section II provides a brief introduction of the unsupervised domain adaptation problem. Section III presents the proposed CDA network in detail. Experimental results are discussed in Section IV, and the conclusion is drawn in Section V.

II. Unsupervised Domain Adaptation Problem

Unsupervised domain adaptation aims to make use of the prior knowledge of the source domain to learn a classifier for the target domain where label information is unavailable. Classification results may not be satisfactory if the classifier trained on the source data is directly used to deal with the target data, because data distributions between the two domains may be different.

In this article, the class-wise adversarial adaptation approach and the PMMD strategy are employed in cooperation with the neural networks. The collected N_s instances from the source domain are denoted as $\mathbf{X}_s \in R^{D\times N_s}$ with class labels $\mathbf{Y}_s \in R^{1\times N_s}$, where D is the dimensionality of the data. The collected N_t unlabeled instances from the target domain are denoted as $\mathbf{X}_t \in R^{D\times N_t}$. The number of classes is C for both domains. For adversarial learning, let $\mathbf{D}_s \in R^{1\times N_s}$ and $\mathbf{D}_t \in R^{1\times N_t}$ denote the domain labels of the source data and the target data, respectively, with the elements of $\mathbf{D}_s$ equaling to 0 and the elements of $\mathbf{D}_t$ equaling to 1.

III. Proposed Domain Adaptation Method

The proposed domain adaptation method is implemented with a neural network, where the class-wise adversarial adaptation and the class-wise PMMD adaptation are combined to obtain the invariant feature representation. The flowchart of the CDA is shown in Fig. 1, which contains three parts: a feature extractor G_f, a classifier G_c, and C

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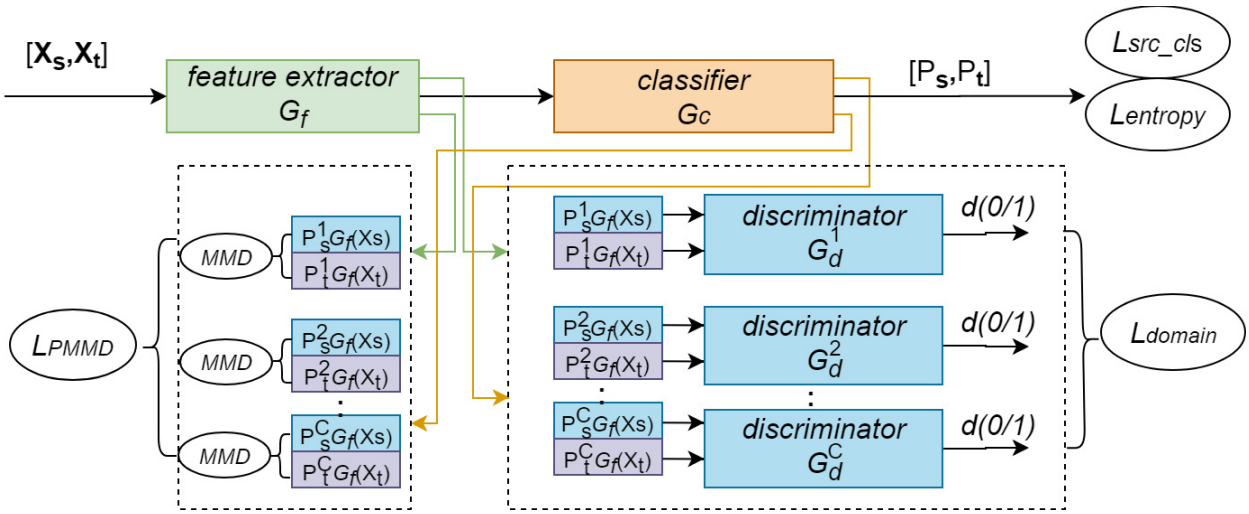


Fig. 1. Flowchart of the CDA algorithm.

discriminators $G_d^c|_{c=1}^C$. The feature extractor G_f generates domain-invariant features $G_f(\mathbf{X}_s)$ and $G_f(\mathbf{X}_t)$ for the source data and the target data, respectively. The classifier G_c outputs the probability-prediction results $\mathbf{P}_s \in R^{C\times Ns}$ and $\mathbf{P}_t \in R^{C\times Nt}$, and the domain discriminator G_d^c yields the domain predictions for data from the cth class. In addition, the MMD constraint is included to contribute to the training of the feature extractor. It is worth noting that the CDA network is only composed of full-connected layers for the pixel-level classification task.

The loss function of the CDA is defined as

$$\begin{aligned} &L\left(\mathbf{X}_s, \mathbf{Y}_s, \mathbf{X}_t, \mathbf{D}_s, \mathbf{D}_t; \boldsymbol{\theta}_f, \boldsymbol{\theta}_d^c|_{c=1}^C, \boldsymbol{\theta}_c\right) \\ &= L_{\text{src_cls}}\left(\mathbf{X}_s, \mathbf{Y}_s; \boldsymbol{\theta}_f, \boldsymbol{\theta}_c\right) \\ &\quad - \lambda_1 L_{\text{domain}}\left(\mathbf{X}_s, \mathbf{D}_s, \mathbf{X}_t, \mathbf{D}_t; \boldsymbol{\theta}_f, \boldsymbol{\theta}_d^c|_{c=1}^C\right) \\ &\quad + \lambda_2 L_{\text{PMMD}}\left(\mathbf{X}_s, \mathbf{X}_t; \boldsymbol{\theta}_f\right) \\ &\quad + \beta L_{\text{entropy}}\left(\mathbf{X}_t; \boldsymbol{\theta}_f, \boldsymbol{\theta}_c\right) \end{aligned} \tag{1}$$

where the first term denotes the classification loss on the source labeled data, the second term represents the domain classification loss of the discriminators, the third term expresses the PMMD loss, and the fourth term denotes the entropy constraint on the classification results of the target data. The notations λ_1, λ_2, and β are the tradeoff hyperparameters, $\boldsymbol{\theta}_f$ represents the parameter of the feature extractor G_f, $\boldsymbol{\theta}_c$ denotes the parameter of the classifier G_c, and $\boldsymbol{\theta}_d^c|_{c=1}^C$ is the parameter of the cth domain discriminator.

In the CDA network, the feature extractor G_f is trained by all the four losses, the classifier G_c is updated by the classification loss of the source labeled instances $L_{\text{src_cls}}$ and the entropy regularization loss L_{entropy}, and the domain discriminator $G_d^c|_{c=1}^C$ is learned by the domain classification loss L_{domain}. The object of the whole loss function is to find the network parameters $\boldsymbol{\theta}_f^*, \boldsymbol{\theta}_c^*, \boldsymbol{\theta}_d^{c*}|_{c=1}^C$ that obey

$$\left(\boldsymbol{\theta}_f^*, \boldsymbol{\theta}_c^*\right) = \arg\min_{\boldsymbol{\theta}_c, \boldsymbol{\theta}_f} L\left(\mathbf{X}_s, \mathbf{Y}_s, \mathbf{X}_t, \mathbf{D}_s, \mathbf{D}_t; \boldsymbol{\theta}_f, \boldsymbol{\theta}_d^c|_{c=1}^C, \boldsymbol{\theta}_c\right) \tag{2}$$

$$\begin{aligned} &\left(\boldsymbol{\theta}_d^{1*}, \ldots, \boldsymbol{\theta}_d^{C*}\right) \\ &= \arg\max_{\boldsymbol{\theta}_d^c|_{c=1}^C} L\left(\mathbf{X}_s, \mathbf{Y}_s, \mathbf{X}_t, \mathbf{D}_s, \mathbf{D}_t; \boldsymbol{\theta}_f, \boldsymbol{\theta}_d^c|_{c=1}^C, \boldsymbol{\theta}_c\right). \end{aligned} \tag{3}$$

A. Source Classification Loss

In the CDA network, source-labeled data are used to train the classifier G_c. Since $G_f(\mathbf{X}_s)$ and $G_f(\mathbf{X}_t)$ are the domain-invariant features, the classifier G_c trained on the source data $G_f(\mathbf{X}_s)$ can be directly used for the classification of the target data $G_f(\mathbf{X}_t)$ in the common feature space. The classification loss on the source-labeled data also participates in training the feature extractor G_f to be more discriminative.

For the multicategory classification task in supervised learning, cross-entropy loss is often used in the deep network. The cross-entropy loss of a source-labeled instance $\mathbf{x}$ is defined as

$$L_y(\mathbf{p}, \mathbf{y}) = -\sum_{c=1}^{C} y^c \log p^c \tag{4}$$

$$p^c = S\left(G_c\left(G_f(\mathbf{x})\right)_c\right) = \frac{\exp\left(G_c\left(G_f(\mathbf{x})\right)_c\right)}{\sum_{i=1}^{C} \exp\left(G_c\left(G_f(\mathbf{x})\right)_i\right)} \tag{5}$$

where $\mathbf{y}$ is the one-hot encoding of the label information of point $\mathbf{x}$, $\mathbf{p}$ is the predicted probability output obtained by classifier G_c, and p^c denotes the probability of $\mathbf{x}$ belonging to the cth class. The $G_c(G_f(\mathbf{x}))_c$ represents the prediction value of the cth node. The $S(\cdot)$ denotes the softmax function. Therefore, the classification loss of all the source instances is defined as

$$\begin{aligned} &L_{\text{src_cls}}\left(\mathbf{X}_s, \mathbf{Y}_s; \boldsymbol{\theta}_f, \boldsymbol{\theta}_c\right) \\ &\quad = \frac{1}{N_s} \sum_{(\mathbf{x}_i, \mathbf{y}_i)\sim(\mathbf{X}_s, \mathbf{Y}_s)} L_y\left(S\left(G_c\left(G_f(\mathbf{x}_i)\right)\right), \mathbf{y}_i\right). \end{aligned} \tag{6}$$

B. Class-Wise Adversarial Adaptation

The learning of the feature extractor and the discriminator is adversarial, since the discriminator aims to distinguish the source domain and the target domain, while the feature extractor attempts to make confusion across the domains. In the original adversarial adaptation network, the domain discriminator performs a two-category classification, which regards all the source data as a class and all the target data as the other class [21], [31]. The marginal distribution shift

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across the domains can be reduced by maximizing the domain classification error for the feature extractor and minimizing the error for the discriminator. However, different classes may have different spectral drifts. The marginal distribution adaptation cannot guarantee the distribution adaptation of each specific class. Therefore, C domain discriminators $G_d^c|_{c=1}^C$ should be designed to align the data distributions of each class, resulting in class-wise adversarial adaptation.

As shown in Fig. 1, each discriminator is responsible for matching one associated class between the two domains. For the cth domain discriminator, the inputs are the probability-weighted features $\mathrm{P}_s^c G_f(\mathbf{X}_s)$ and $\mathrm{P}_t^c G_f(\mathbf{X}_t)$, where P_s^c and P_t^c are obtained by the classifier G_c, and $G_f(\mathbf{X}_s)$ and $G_f(\mathbf{X}_t)$ are generated by the feature extractor. P_s^c and P_t^c characterize the probability of the source data and the target data belonging to the cth class, respectively, and thus, they are used as the weights to determine how much each sample should be sent to the cth domain discriminator G_d^c. The probability-weighted feature can mitigate the problem of hard assignment of each instance to only one domain discriminator.

The domain classification loss for all the data is defined as

$$L_{\text{domain}}\left(\mathbf{X}_s, \mathbf{D}_s, \mathbf{X}_t, \mathbf{D}_t; \boldsymbol{\theta}_f, \boldsymbol{\theta}_d^c|_{c=1}^C\right) = \frac{1}{N_s + N_t} \sum_{c=1}^{C} \sum_{(\mathbf{x}_i, d_i) \in (\mathbf{X}_s \cup \mathbf{X}_t, \mathbf{D}_s \cup \mathbf{D}_s)} L_d\left(G_d^c\left(p_i^c G_f(\mathbf{x}_i)\right), d_i\right) \quad (7)$$

$$L_d\left(h_i^c, d_i\right) = -d_i \log h_i^c - (1 - d_i) \log\left(1 - h_i^c\right) \quad (8)$$

where d_i is the domain label of the data $\mathbf{x}_i$, p_i^c denotes the probability of assigning $\mathbf{x}_i$ to the cth class, $p_i^c G_f(\mathbf{x}_i)$ represents the probability-weighted feature of $\mathbf{x}_i$, and $G_d^c(p_i^c G_f(\mathbf{x}_i))$ denotes the domain prediction of the cth discriminator. If p_i^c is large, which means it has a high probability to be from the cth class, $G_f(\mathbf{x}_i)$ will go to the cth discriminator G_d^c and contribute to the domain confusion of the cth class with a high probability. L_d is the domain classification loss function on a data point $\mathbf{x}_i$, where h_i^c represents the predicted domain from the cth discriminator.

Then, we have the objective function of using the domain classification loss to train the feature extractor and the discriminator

$$\left(\hat{\boldsymbol{\theta}}_f^*\right) = \arg\max_{\boldsymbol{\theta}_f} L_{\text{domain}}\left(\mathbf{X}_s, \mathbf{D}_s, \mathbf{X}_t, \mathbf{D}_t; \boldsymbol{\theta}_f, \boldsymbol{\theta}_d^c|_{c=1}^C\right) \quad (9)$$

$$\left(\boldsymbol{\theta}_d^{1*}, \ldots, \boldsymbol{\theta}_d^{C*}\right) = \arg\min_{\boldsymbol{\theta}_d^c|_{c=1}^C} L_{\text{domain}}\left(\mathbf{X}_s, \mathbf{D}_s, \mathbf{X}_t, \mathbf{D}_t; \boldsymbol{\theta}_f, \boldsymbol{\theta}_d^c|_{c=1}^C\right). \quad (10)$$

It is worth noting that the feature extractor is determined by all the four losses in (1), so we denoted the parameters of the feature extractor trained by the domain classification loss as $\hat{\boldsymbol{\theta}}_f^*$.

The discriminator aims to distinguish the source domain and the target domain, and thus, the domain classification loss should be minimized. For class-wise adaptation, we have C discriminators, where each of them focuses on separating the two domains of one class. The feature extractor attempts to generate features that cannot be distinguished by all the discriminators, and therefore, the domain classification loss should be maximized. Through the adversarial learning between the discriminators and the feature extractor, the source-like target features and the target-like source features are generated. For each class c, the features $G_f(\mathbf{X}_s)$ and $G_f(\mathbf{X}_t)$ are not distinguishable, and therefore, the conditional distribution adaptation can be achieved.

Using the standard backpropagation algorithm, the saddle points of parameters $\boldsymbol{\theta}_f, \boldsymbol{\theta}_d^c|_{c=1}^C$ can be reached. To maximize the domain classification error for the feature extractor and minimize the domain classification loss for C discriminators simultaneously, the gradient reversal layer (GRL) is designed between the feature extractor and the discriminator. The GRL performs an identity transform in the forward pass and acts as a gradient-reversal operation in the backward propagation procedure [31]. In the neural networks, gradient reversal can be easily performed, which multiplies the backpropagation function by a negative one. With the application of GRL, we do not have to train alternately the feature extractor and the discriminator, but accomplish the training of them in one step.

C. Probability Maximum Mean Discrepancy

In the domain adaptation methods, MMD is a commonly used metric of discrepancy between two distributions because of its efficiency in computation and optimization [37]. It can be conducted on each class to achieve conditional distribution alignment. Since there is no labeled information in the target domain, the probability predictions of the target data are used to estimate the means of each class. The loss function of the class-wise PMMD is defined as

$$L_{\text{PMMD}}\left(\mathbf{X}_s, \mathbf{X}_t; \boldsymbol{\theta}_f\right) = \sum_{c=1}^{C} \left\| \frac{1}{N_s} \sum_{x_i \in \mathbf{X}_s} p_i^c G_f(\mathbf{x}_i) - \frac{1}{N_t} \sum_{x_j \in \mathbf{X}_t} p_j^c G_f\left(\mathbf{x}_j\right) \right\|_2^2. \quad (11)$$

By minimizing the PMMD, the source data and the target data become similar on the category level. In the traditional MMD-based domain adaptation approaches, the MMD can be calculated by singular value decomposition [38]. Differently, deep network optimization generally makes use of the mini-batch-based backpropagation method to find a locally optimal solution rather than an analytic solution. Fortunately, Zhang *et al.* [39] achieved domain adaptation by embedding the MMD into a deep network and proved that the mini-batches can be sent into a network to achieve the MMD match, which is equivalent to matching the whole data set.

D. Entropy Regularization

Both adversarial adaptation and PMMD perform per-class alignment to avoid negative transfer. However, the distribution estimation of each target class may not be accurate due to the lack of labeled information in the target domain. Since adaptation relies on the probability-prediction results, we apply a constraint to the probability predictions to guarantee a low-density separation between the classes, which means a peak distribution of the probability-prediction output rather than a smooth distribution is more desired. Minimizing the

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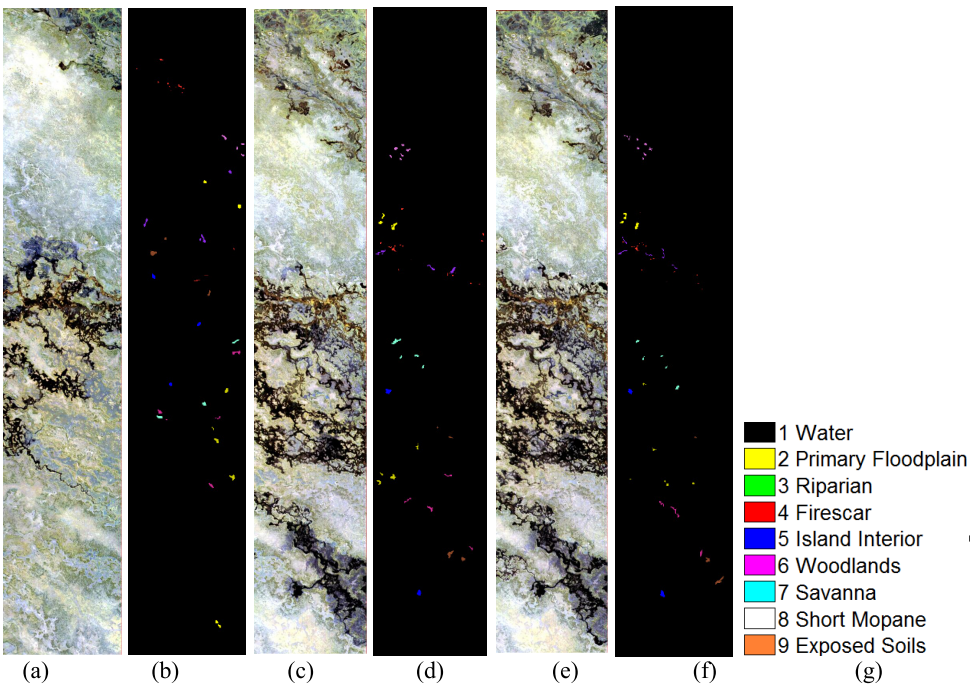


Fig. 2. Hyperion images of BOT in (a) May, (c) June, and (e) July. Ground truth of (b) May image, (d), and (f) July image. (g) Class legend.

entropy of the probability outputs can achieve this purpose [40], and the entropy regularization is defined as

$$L_{\text{entropy}}\left(\mathbf{X}_t;\boldsymbol{\theta}_f,\boldsymbol{\theta}_c\right)=-\frac{1}{N_t}\sum_{\mathbf{x}_i\in\mathbf{X}_t}\sum_{c=1}^{C}p_i^c\log p_i^c. \tag{12}$$

It is worth noting that the entropy regularization is not applied on the source data, since their probability-prediction results can well obey a peak distribution after minimizing the classification loss of the source data.

E. Related Works and Discussions

The relationships between the proposed CDA network and several related works are described as follows.

1) Domain Adversarial Neural Network [31]: Ganin *et al.* [31] first applied the adversarial learning to the domain adaptation field and proposed the DANN. The DANN includes a domain classifier, which regards all the source data as a class and all the target data as the other class. The purpose of the adversarial learning is to make the two domains similar and obtain the marginal distribution adaptation. In the proposed CDA approach, we considered the difference between different classes and conducted the adversarial adaptation on each class, and thus, the class-conditional distribution adaptation is achieved.

2) Deep Adaptation Network [18]: The DAN achieved domain adaptation by introducing the multiple-kernel MMD strategy into a deep network. However, it only considered the marginal distribution adaptation and did not consider the conditional distribution adaptation. In the proposed CDA network, we combined the class-wise MMD strategy and the class-wise adversarial adaptation approach to obtain a superior feature-alignment performance.

3) Multiple Adversarial Domain Adaptation [33]: Both the MADA network and the proposed CDA network consider the conditional distribution adaptation by using multiple domain discriminators. The differences between them include the following.

1) The proposed CDA combined the class-wise MMD strategy that is more robust to the false pseudo-labels.
2) The CDA used an entropy regularization on the target data to obtain a peak distribution of the probability predictions.
3) The CDA is applied to hyperspectral remote sensing image classification, while the MADA is used for visual domain adaptation.

IV. Experimental Results and Analysis

A. Data Description

Images from two remote sensing sensors were used for experiments. The first one was collected by the Earth Observation satellite-1 (EO-1) Hyperion instrument in May, June, and July 2001 over the Okavango Delta, Botswana (BOT). This image contains 242-band data at a 30-m spatial resolution, covering 357–2576-nm spectrum in 10-nm bands. Uncalibrated and noisy bands were removed, and 145 bands remained. The three images contain nine identified landcover types. The pseudo-color image and label information in May, June, and July are shown in Fig. 2. Any two of the three images can be selected as two domains, and therefore, six data pairs can be used for experiments. The class names and number of samples in each class of the three images are listed in Table I.

The second data were collected using the NASA Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) instrument on March 23, 1996 at an 18-m spatial resolution over the area of Kennedy Space Center (KSC) in Florida. It acquires

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TABLE I
CLASS NAMES AND NUMBER OF SAMPLES OF KSC AND BOT DATA

KSC				BOT				
ID	Class	KSC1	KSC3	ID	Class	May	June	July
1	Scrub	761	422	1	Water	297	361	185
2	Willow Swamp	243	180	2	Primary Floodplain	437	308	96
3	Cabbage Palm Hammock	256	431	3	Riparian	448	303	164
4	Cabbage Palm/Oak	252	132	4	Firescar	354	335	186
5	Slash Pine	161	166	5	Island Interior	337	370	131
6	Oak/Broadleaf Hammock	229	274	6	Woodlands	357	324	169
7	Hardwood Swamp	105	248	7	Savanna	330	342	171
8	Graminoid Marsh	431	453	8	Short Mopane	239	299	152
9	Salt Marsh	419	156	9	Exposed Soils	215	229	96
10	Water	927	1392					



(a) (b) (c) (d) (e)

Fig. 3. RGB images of (a) KSC1 and (c) KSC3. Ground references of (b) KSC1 and (d) KSC3. (e) Class legend.

224-band data that cover 400–2500-nm portion of the spectrum. After removing water absorption and noisy bands, 176 bands remained. Two spatially disjoint images (KSC1 and KSC3) contain ten common landcover types. The pseudocolor images and label information are shown in Fig. 3. The class names and the number of samples in each class of the two images are listed in Table I.

We chose eight data pairs for experiments. For BOT May, June, and July images, six data pairs were used, which are denoted as May–June, June–May, May–July, July–May, June–July, and July–June. In each data pair, the first data set denotes the source domain and the second denotes the target domain. For KSC hyperspectral images, two data pairs (KSC1-KSC3 and KSC3-KSC1) were used.

B. Comparison With Other Domain Adaptation Methods

The proposed domain adaptation network was compared with four popular deep transfer network domain adaptation methods, including DAN [18], D-CORAL [19], DANN [31], and MADA [33]. In addition, a deep network with the rectified linear unit (ReLU) activation function which is denoted as RDNN was also employed for comparison [20]. It directly uses the source data as training data to classify the target data.

In the proposed CDA, the feature extractor includes two hidden layers (250 units for BOT and 256 units for KSC), and the classifier is composed of a softmax output layer with dimensionality being set to C. Each of the domain discriminators contains one hidden layer (100 units) that conducts binary domain classification. The ReLU-activation function was employed in the hidden layers. The network structures of the RDNN, DAN, and D-CORAL are the same, which contains a feature extractor G_f and a classifier G_c, and the hyperparameters of G_f and G_c are the same as those of CDA. The DANN contains a feature extractor G_f, a classifier G_c, and one domain discriminator G_d, and their hyperparameters are the same as those of CDA. The structures of CDA and MADA are very similar, and the difference between them is that the former includes the PMMD part and entropy regularization, while the latter does not.

Before the network training, each spectral band was normalized to have standard normal distribution $N(0, 1)$ for both source and target data. Since both domains have the same mean values, this preprocessing method is useful to reduce the overall distribution shift across the domains. For a fair comparison, such a data-processing method was applied to all the compared methods. It is worth noting that we also conducted an RDNN with another preprocessing method, which scales the whole data set to $[-1, 1]$. Since this normalization does not have the domain adaptation effect like performing $N(0, 1)$ on each spectral band, the RDNN with this processing method was named as RDNN with none of adaptation and denoted as RDNN(NA).

In the CDA network, the batch size was set to be 128, and the mini-batch stochastic gradient descent (SGD) was used with a momentum of 0.9 and a weight decay of 0.0001. The learning rate was set to be $\eta = \eta_0/(1 + \alpha i)^p$, where α was set to be 10, p was set to be 0.75, the training step i changed from 0 to 1, and the initial learning rate η_0 was set to be 0.005 for the BOT data and KSC1-KSC3, and 0.0025 for KSC3-KSC1. Moreover, the number of epochs was 1000 for the BOT data and 600 for the KSC data. For the tradeoff parameters λ_1 and λ_2, we first introduced a parameter $\lambda = 2/(1 + \exp(-\delta i)) - 1$,

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TABLE II
OA% OF DIFFERENT DOMAIN ADAPTATION METHODS

Data sets		RDNN(NA)	RDNN	DAN	D-CORAL	DANN	MADA	**CDA**
BOT	May–June	72.30	89.38	89.80	90.44	87.83	91.51	**92.62**
	June–May	62.84	80.11	83.57	84.53	78.30	82.12	**87.85**
	May–July	66.53	86.00	87.52	87.87	86.52	90.29	**91.27**
	July–May	57.07	68.47	72.97	74.97	69.04	73.10	**79.78**
	June–July	88.72	92.86	91.79	92.14	92.90	94.76	**95.05**
	July–June	84.20	89.63	90.26	89.97	89.84	91.54	**93.14**
KSC	KSC1–KSC3	61.32	67.57	68.11	67.10	70.32	72.99	**74.21**
	KSC3–KSC1	64.42	62.70	62.52	67.08	65.03	74.69	**76.20**

TABLE III
KAPPA COEFFICIENTS OF DIFFERENT DOMAIN ADAPTATION METHODS

Data sets		RDNN(NA)	RDNN	DAN	D-CORAL	DANN	MADA	CDA
BOT	May–June	0.686	0.880	0.885	0.892	0.863	0.904	**0.916**
	June–May	0.582	0.776	0.814	0.825	0.755	0.798	**0.876**
	May–July	0.621	0.841	0.859	0.862	0.847	0.890	**0.902**
	July–May	0.518	0.645	0.695	0.717	0.652	0.696	**0.772**
	June–July	0.872	0.919	0.907	0.911	0.919	0.941	**0.944**
	July–June	0.821	0.883	0.890	0.887	0.885	0.908	**0.923**
KSC	KSC1–KSC3	0.530	0.604	0.610	0.597	0.638	0.670	**0.684**
	KSC3–KSC1	0.592	0.571	0.557	0.618	0.589	0.704	**0.722**

where δ was set to be 10. The value of λ is close to zero during the early training (i is small) and becomes large (less than 1) gradually . This parameter setting strategy can stabilize parameter sensitivity, since it allows the domain classifier to be less sensitive to the noisy signal at the early stages of the training procedure [31]. Next, we tested different ratios of λ_1 and λ_2. When the ratios were set to be 20:1, 15:1, 10:1, 5:1, 2:1, and 1:1, we fixed $\lambda_1 = \lambda$; when the ratios were 1:2, 1:5, 1:10, 1:15, and 1:20, $\lambda_2 = \lambda$ was adopted. This parameter setting method can guarantee the maximum value of λ_1 and λ_2 to be less than 1. For parameter β, we selected the values in the range [0.05, 0.15, 0.25, 0.35, 0.5, 0.7, 1.0]. Since the initial network parameters were chosen randomly, experiments were conducted ten times and the average classification accuracy was used for evaluation. For the comparative methods, the DANN and MADA have a parameter for the domain classification loss, the DAN includes a weight of the multiple-kernel MMD loss, and the D-CORAL contains a weight of the CORAL loss. These parameters were chosen as the recommended values in [18], [19], [31], and [33].

The overall accuracy (OA) and kappa coefficients of these algorithms on the eight data sets are listed in Tables II and III. It can be seen that the RDNN outperformed the RDNN(NA) for most of the data sets, which demonstrates that performing normal initialization $N(0, 1)$ on each spectral band is useful for domain adaptation. If the spectral shift across the domains is small (for example, BOT June-July), the RDNN can achieve a satisfactory performance. However, for the data pairs that have big spectral drift (such as BOT July–May, KSC1-KSC3, and KSC3-KSC1), the accuracies of the RDNN are low. Almost all the domain adaptation methods obtain higher accuracies than the RDNN, demonstrating their positive transfer learning ability. The DANN can obtain higher accuracies than the RDNN for most of the data sets. The MADA outperforms the RDNN and DANN, because MADA considers the alignment of each specific class. The CDA can yield the best performances for all data sets. It can obtain significant improvements compared with the RDNN and achieve better performances than other domain adaptation networks, which demonstrates that the adversarial adaptation, PMMD strategy, and entropy regularization can cooperate well for the domain adaptation of the hyperspectral remote sensing images.

C. Alignment Performance of Our Adaptation Method

The proposed CDA method aims to extract domain-invariant features. This experiment was used to show the alignment performance of our adaptation method. Fig. 4 plots the features.

The features from the 10th and 150th units of the feature extractor were used for illustrating the alignment performance

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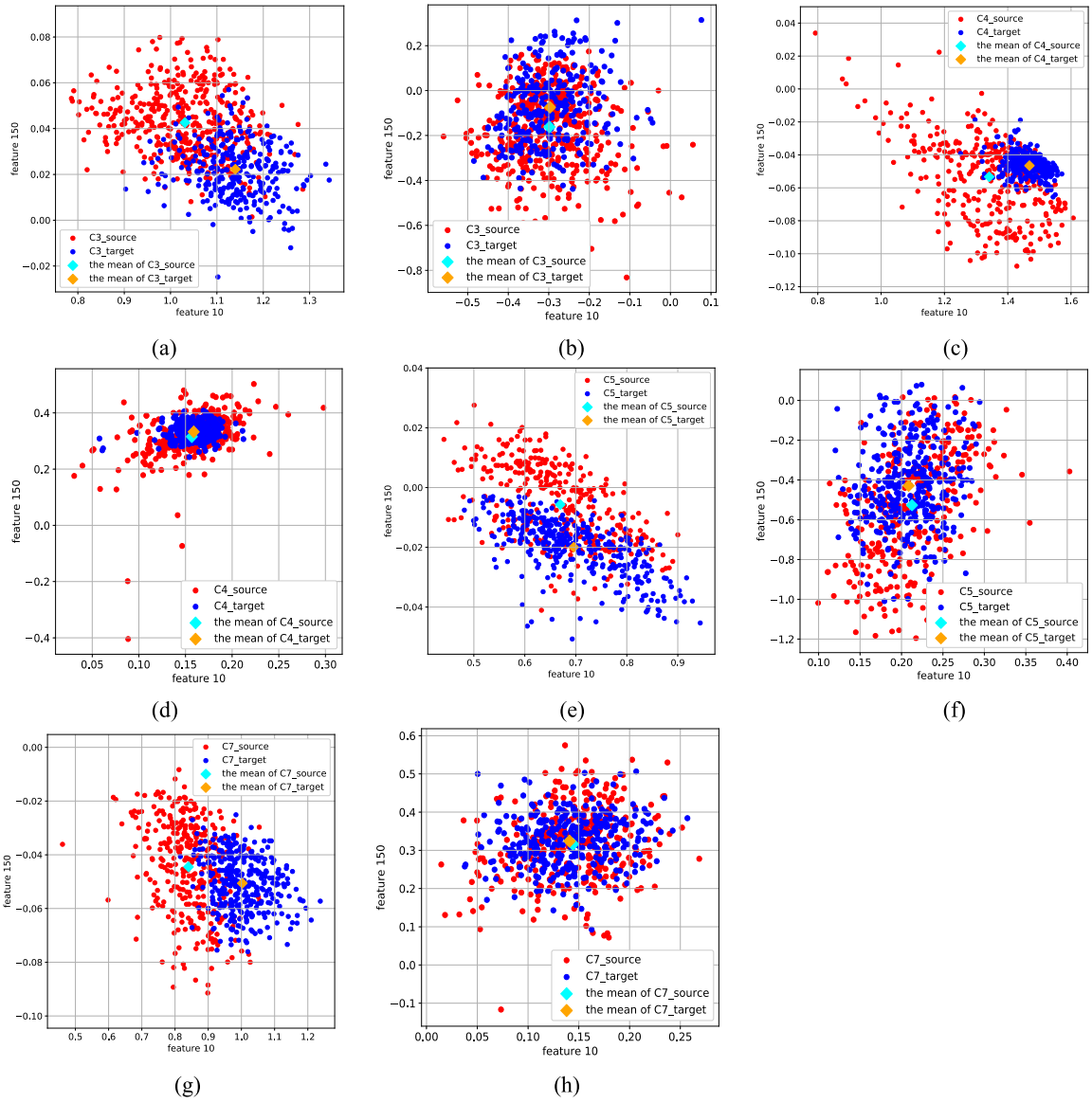


Fig. 4. Alignment performance of the BOT May–June data. (a) Class 3 with RDNN(NA). (b) Class 3 with CDA. (c) Class 4 with RDNN(NA). (d) Class 4 with CDA. (e) Class 5 with RDNN(NA). (f) Class 5 with CDA. (g) Class 7 with RDNN(NA). (h) Class 7 with CDA.

of the BOT May–June data. Points from different domains were plotted with different colors, and the class means were represented by the bigger orange and green rhombus. We selected classes 3, 4, 5, and 7 for illustration. Fig. 4(a), (c), (e), and (g) shows the features extracted by the RDNN(NA). The distribution differences can be observed, since the RDNN(NA) does not apply the domain adaptation strategy. With the CDA approach, the features of each class are better aligned and the centroids of each class become closer, as shown in Fig. 4(b), (d), (f), and (h). This feature visualization intuitively demonstrates the effectiveness of the proposed adaptation method.

We also plotted the features of all classes in one figure. The t-distributed Stochastic Neighbor Embeddings (t-SNE) [41] were used to visualize the intermediate features of the RDNN(NA) and CDA approaches. We used BOT May–June, June–May, and May–July data pairs for illustration. Fig. 5 shows the t-SNE features, where Fig. 5(a)–(c) plots the features extracted by the RDNN(NA) without the adaptation strategy, and Fig. 5(d)–(f) illustrates the features extracted by the CDA. In each figure, different classes were displayed in different colors, and the points in the same class from different domains were plotted with the same color but different shapes. From Fig. 5(a)–(c), it can be observed that some classes have large within-class variances, indicating the feature differences of data from the same class across the domains. In addition, different classes may have overlapping features, which produces difficulty in classification. As shown in Fig. 5(d)–(f), the features of the same class between the domains are well aligned after applying the CDA approach. The approach also provides good separation of different classes.

D. Analysis of Our Approach

The CDA network contains two domain adaptation strategies, class-wise adversarial adaptation and class-wise PMMD, and introduces an entropy regularization to improve

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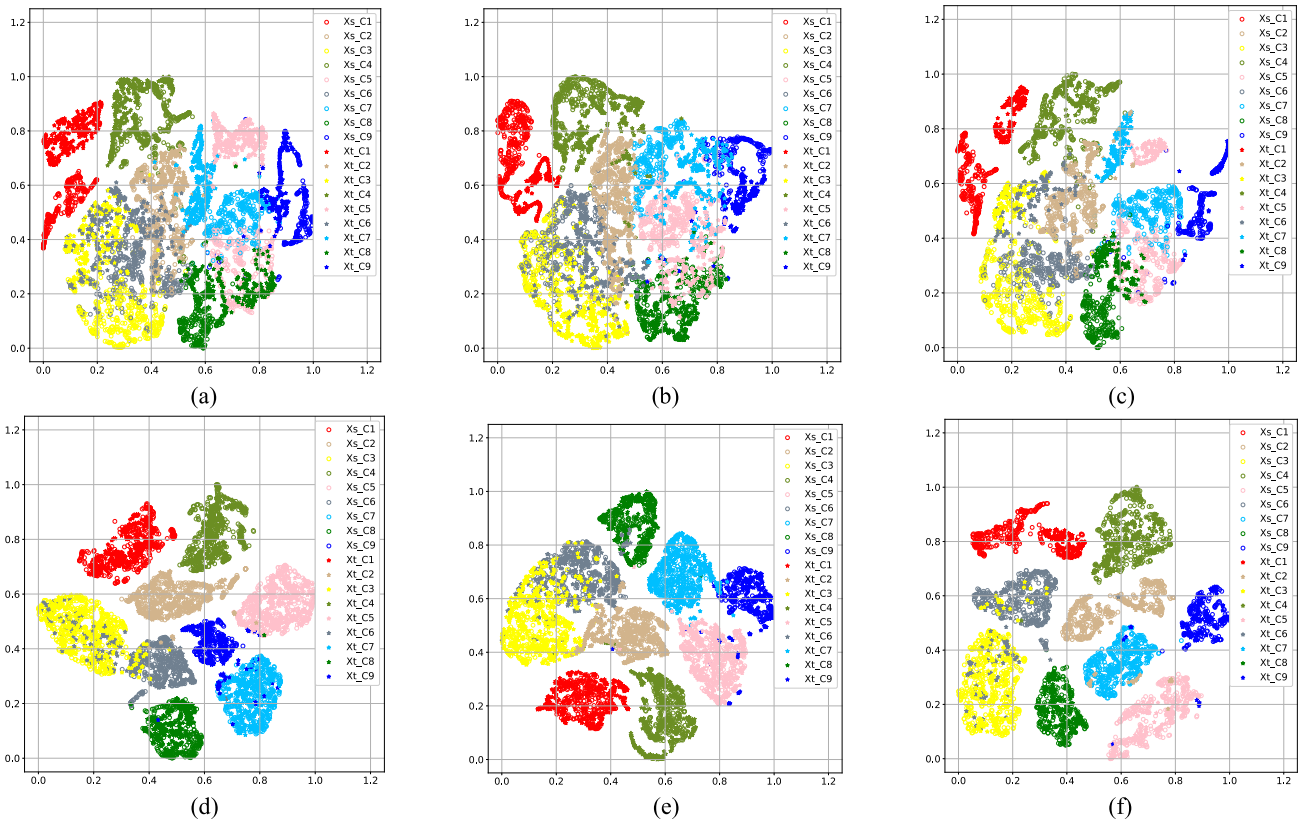


Fig. 5. t-SNE visualization of features extracted by RDNN(NA) and CDA. (a) RDNN(NA) for BOT May–June. (b) RDNN(NA) for BOT June–May. (c) RDNN(NA) for BOT May–July. (d) CDA for BOT May–June. (e) CDA for BOT June–May. (f) CDA for BOT May–July.

TABLE IV
OA% OF THE RELATED DOMAIN ADAPTATION METHODS

Data sets		DANN	PMMD	MADA	CDA (w/o entropy)	**CDA**
BOT	May–June	87.83	91.27	91.51	92.05	**92.62**
	June–May	78.30	86.74	82.12	87.55	**87.85**
	May–July	86.52	89.75	90.29	90.91	**91.27**
	July–May	69.04	76.39	73.10	79.04	**79.78**
	June–July	92.90	93.26	94.76	94.97	**95.05**
	July–June	89.84	90.90	90.90	92.53	**93.14**
KSC	KSC1–KSC3	70.32	71.06	72.99	73.37	**74.21**
	KSC3–KSC1	65.03	75.92	74.69	**76.46**	76.20

the predictions. This experiment focused on analyzing these terms by comparing the CDA network with another four networks (DANN, PMMD, MADA, and CDA without entropy), where the DANN only applied one discriminator to achieve marginal distribution adaptation, the PMMD denotes a transfer network that used the PMMD alignment strategy, and the MADA employed class-wise adversarial adaptation. To observe the effectiveness of applying the entropy constraint, we conducted CDA (w/o entropy) for comparison.

The OA of the five algorithms on the eight data pairs was listed in Table IV, and the kappa coefficient was listed in Table V. Several observations can be obtained.

1) MADA, CDA (w/o entropy), and CDA outperformed the DANN, which demonstrates the effectiveness of class-wise adversarial adaptation compared with the marginal distribution adaptation.
2) CDA (w/o entropy) and CDA obtained higher accuracies than the PMMD and MADA, indicating the value of combining two different domain adaptation strategies.
3) CDA outperformed CDA (w/o entropy) for most of the data pairs, demonstrating the value of entropy regularization, which improves the reliability of the predicted labels.

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TABLE V
KAPPA COEFFICIENT OF THE RELATED DOMAIN ADAPTATION METHODS

Data sets		DANN	PMMD	MADA	CDA (w/o entropy)	**CDA**
BOT	May–June	0.863	0.901	0.904	0.910	**0.916**
	June–May	0.755	0.850	0.798	0.859	**0.876**
	May–July	0.847	0.884	0.890	0.897	**0.902**
	July–May	0.652	0.733	0.696	0.763	**0.772**
	June–July	0.919	0.924	0.941	0.942	**0.944**
	July–June	0.885	0.897	0.897	0.914	**0.923**
KSC	KSC1–KSC3	0.638	0.646	0.670	0.674	**0.684**
	KSC3–KSC1	0.589	0.720	0.704	**0.725**	0.722

TABLE VI
CLASSIFICATION ACCURACY OF EACH CLASS OF BOT JULY-JUNE DATA

ID	Class	DANN	PMMD	MADA	CDA (w/o entropy)	**CDA**
1	Water	100.00	100.00	100.00	100.00	**100.00**
2	Primary Floodplain	93.99	85.03	99.02	97.95	**99.42**
3	Riparian	64.62	**68.94**	63.43	65.84	63.86
4	Firescar	93.91	**99.34**	98.92	99.13	99.13
5	Island Interior	92.89	92.30	90.73	95.19	**96.62**
6	Woodlands	77.25	78.76	79.44	79.50	**79.81**
7	Savanna	95.96	99.65	96.98	99.38	**99.88**
8	Short Mopane	97.96	99.66	99.96	100.00	**100.00**
9	Exposed Soils	88.60	91.57	91.57	93.27	**98.04**
	OA%	89.84	90.90	90.90	92.53	**93.14**
	AA%	89.47	90.58	90.47	92.25	**92.98**
	Kappa	0.885	0.897	0.897	0.914	**0.923**

Besides the OA, per-class classification accuracies were also reported for comparison. The BOT July–June data were selected, and the result is shown in Table VI. For almost all the classes, CDA obtained the highest accuracies. The OA, average accuracy (AA), and Kappa coefficient also indicate that the CDA performs better than the other approaches. Moreover, PMMD, MADA, CDA (w/o entropy), and CDA outperform the DANN for all classes, which demonstrates the advantage of using the class conditional distribution adaptation.

E. Sensitivity Analysis of Parameters in CDA Network

We conducted sensitivity analysis of the three parameters (λ_1, λ_2, and β) in the CDA network, where parameter λ_1 controls the weight of the class-wise adversarial adaptation strategy, parameter λ_2 controls the weight of the PMMD loss, and the parameter β denotes the weight of the entropy regularization.

For parameters λ_1 and λ_2, 11 different ratios (20:1, 15:1, 10:1, 5:1, 2:1, 1:1, 1:2, 1:5, 1:10, 1:15, and 1:20) were tested with the value of β equaling to zero. The classification results on the six BOT data pairs were shown in Fig. 6(a). When λ_1 is equal to or is larger than λ_2, classification accuracies are higher. Thus, we suggest that the value of λ_1:λ_2 should be equal to or larger than 1. For parameters β, seven different values (0.05, 0.15, 0.25, 0.35, 0.5, 0.7, and 1.0) were tested with a fixed ratio of λ_1: λ_2 that yields the highest accuracies. As shown in Fig. 6(b), for BOT May–July and June–July data, the CDA method is insensitive to β, while for the other data pairs, a small value of β is preferred. We suggested that the parameter β should not be more than 0.5.

F. Classification Results of the Whole Image by CDA Network

For the BOT data sets, we selected June–May and June–July to illustrate the classification performance on the whole image. Fig. 7(a) shows the RDNN(NA) result without domain adaptation for the BOT June–May data, and Fig. 7(b) shows the CDA result for the BOT June–May data set. Because there is no ground truth for the whole image, the "reference" was obtained by a deep network trained by the

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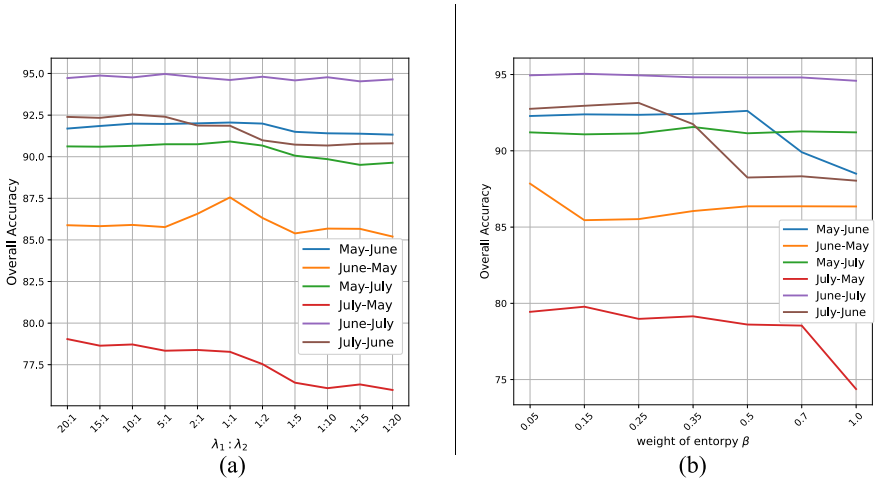


Fig. 6. Sensitivity analysis of parameters in the CDA method using the BOT data sets. (a) Parameter ratio λ_1:λ_2. (b) Parameter β.

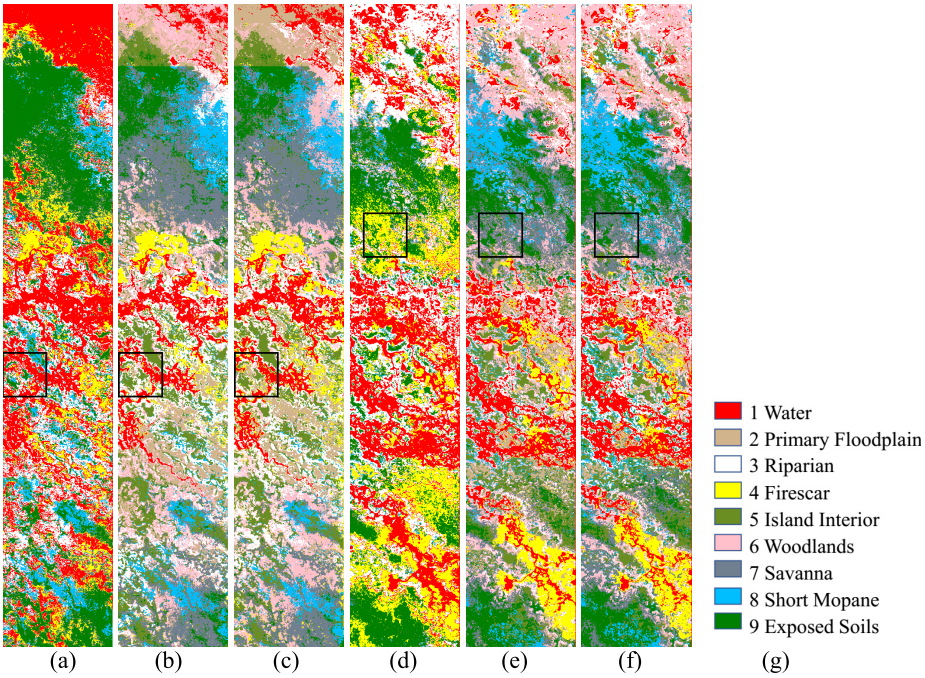


Fig. 7. Classification results of the target image in the BOT June–May and BOT July–June data sets. (a) RDNN(NA) result for the BOT June–May data. (b) CDA for the BOT June–May data. (c) Reference obtained by using target labeled data as training data for the BOT June–May data. (d) RDNN(NA) result for the BOT July–June data. (e) CDA for the BOT July–June data. (f) Reference obtained by using the target labeled data as training data for the BOT July–June data. (g) Class legend.

target labeled data, as shown in Fig. 7(c). The result of CDA and "reference" is very similar, while the result of RDNN(NA) is quite different to the "reference," which demonstrates the effectiveness of the CDA. Similar observations can also be obtained by the BOT June-July data set.

For further analysis, two local regions were selected, which is shown in the black windows in Fig. 7. The BOT data include two major ecosystems defined by the absence or presence of flooding, i.e., upland and wetland [42]. For the BOT June-May data set, a wetland was chosen, which mainly contains class 1(Water, red), class 2 (Primary Floodplain, tan), and class 5 (Island Interior, olice drab). As shown in Fig. 8(a), the RDNN(NA) misclassified class 5 as class 9 (Exposed Soils, dark green) and class 8 (Short Mopane, dark blue). By comparing Fig. 8(b) and (c), the classification results of CDA are close to the "reference." For the July–June data set, an upland area was selected, which mainly contains class 6 (Woodlands, pink), class 7 (Savanna, slate gray), and class 9 (Exposed Soils, dark green). In Fig. 8(d), the RDNN(NA) yielded false predictions as class 4 (Firescar, yellow). By comparing Fig. 8(e) and (f), the classification result of CDA is satisfactory.

G. Computational Time

The computational time of each compared transfer network was reported in Table VII. All experimental results were

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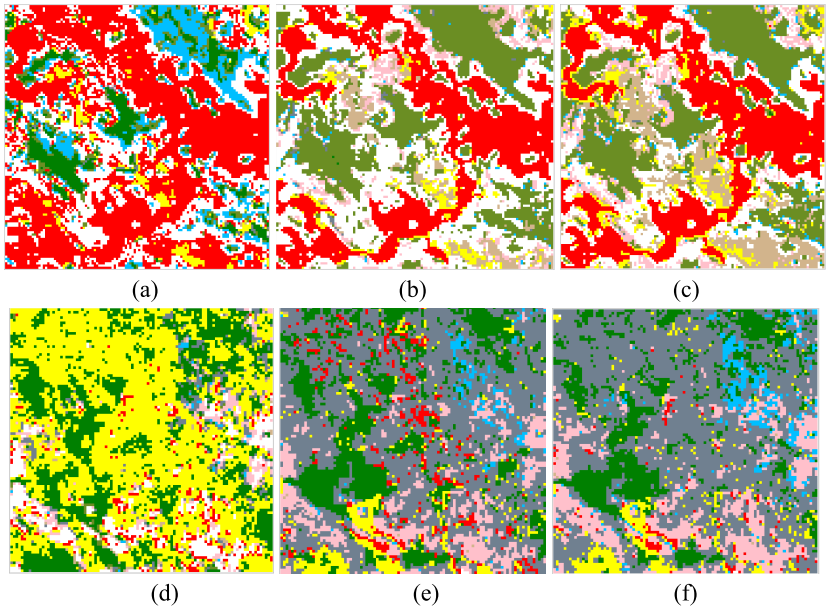


Fig. 8. Classification results of two local regions. (a) RDNN(NA) result for a local region in the BOT June–May data. (b) CDA result for a local region in the BOT June–May data. (c) Reference obtained by using target labeled data as training data for the BOT June–May data. (d) RDNN(NA) result for a local region in the BOT July–June data. (b) CDA result for a local region in the BOT July–June data. (c) Reference obtained by using target labeled data as the training data for the BOT July–June data.

TABLE VII
COMPUTATIONAL TIME (S) OF ONE-STEP TRAINING OF DIFFERENT METHODS

Data sets	RDNN/RDNN(NA)	DAN	D-CORAL	DANN	PMMD	MADA	CDA
BOT	0.010	0.024	0.021	0.027	0.039	0.048	0.065
KSC	0.015	0.025	0.023	0.027	0.034	0.050	0.073

obtained using a machine with an Intel Core i7-8700K Six-core CPU (16-GB RAM) powered with Nvidia GTX 1080Ti GPU with 11-GB memory. The Pytorch deep learning library was used to implement the neural networks of these methods, and the GPU was used to accelerate the training process. We used the computational time of one-step training for comparison, which contains one forward propagation and one backward propagation with a 128-sized batch. As shown in Table VII, the training time of all domain adaptation is longer than the RDNN and the RDNN(NA). In these domain adaptation methods, PMMD and MADA are slower than DAN and DANN, which indicates that aligning the conditional distribution is more time-consuming than aligning the margin distribution. The CDA that combines two domain adaptation strategies costs the most, but the computational time is still acceptable.

V. CONCLUSION

In this article, we proposed an end-to-end unsupervised domain adaptation network based on class-wise adversarial learning. It is able to achieve conditional distribution adaptation by maximizing multiple domain classification errors and minimizing the class-wise PMMD loss simultaneously. In the experiments with hyperspectral remote sensing images, the CDA network obtains superior performance to several popular domain adaptation approaches. Moreover, it outperforms PMMD and MADA, indicating the effectiveness of combining different adaptation strategies. Since the proposed CDA network is composed of fully connection layers, other deep networks (e.g., convolutional neural network) can also be integrated for adaptation purpose. In addition, since one domain discriminator can also achieve CDA, the proposed network may be further simplified and accelerated. Moreover, the adversarial can also cooperate with the band selection [43] to assign weights for different bands. These are our future work.

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